Lecture 20

How to recognize large scale genomic variations

Simulate data of known property

The best way to understand what an analysis does is to generate data of known property then see what you can recover with the analysis.

Many assignments have already asked you to be the "data generator".

We'll do it now on large scale. You'll become the sequencing instrument.

Data simulators

Simulating data allows us to evaluate the performance of our tools and techniques.

Helps you understand the challenges of recognizing real signal from noise.

You can practice on smaller, manageable datasets.

Different biological processes may require different data simulators.

How to be a sequencing instrument

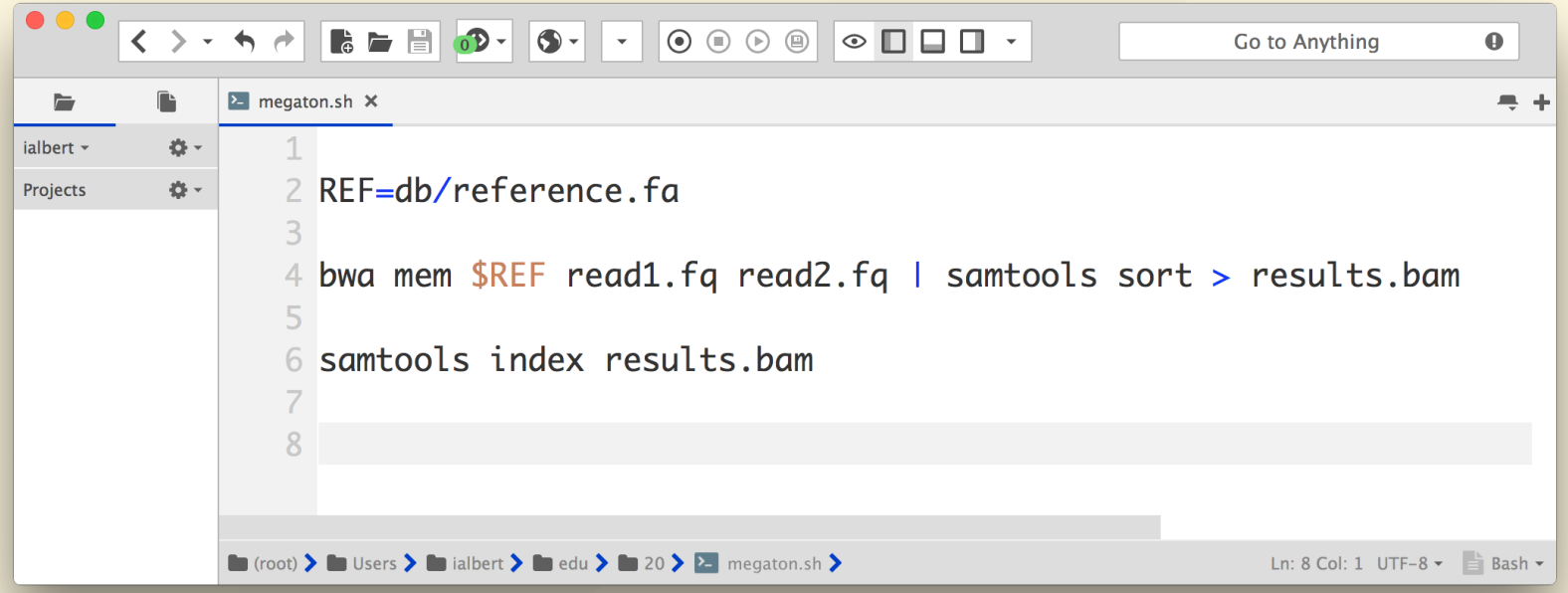
Set up your reference genome:

```
REF=db/reference.fa
mkdir -p db
efetch -db nuccore -id  AF086833 -format fasta > $REF
bwa index $REF
```

A genome simulator generates reads from the reference:

```
wgsim -N 3000 $REF read1.fq read2.fq
```

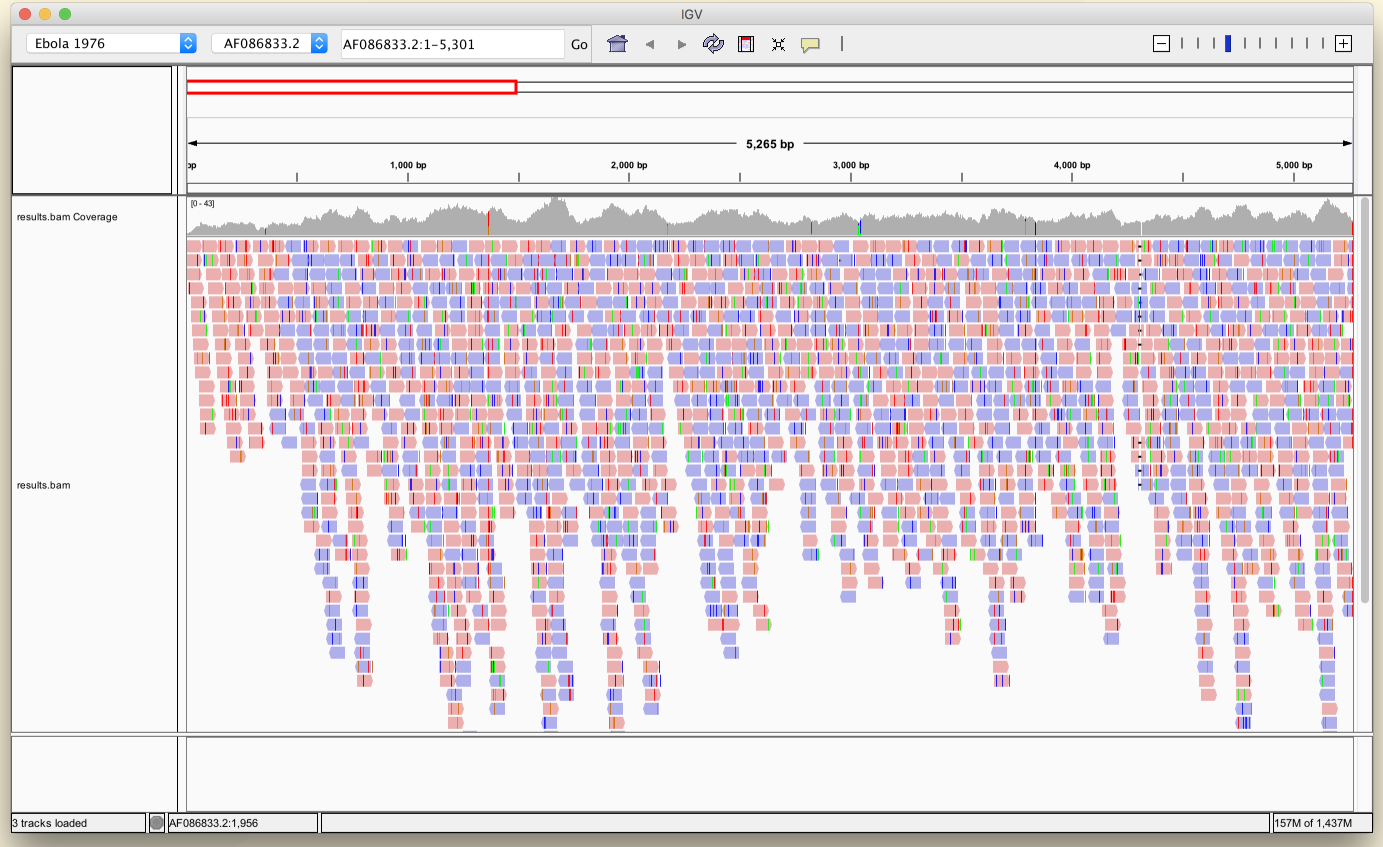
Use scripts, almost a necessity.



Run it with:

```
bash megaton.sh
```

Simulation results



This data was generated from the reference!

Megaton scripts

Liberating the explosive power of the human mind.

I came up this word. I call a **megaton script** a code that automates a tedious tasks so well that it allows your mind to focus and understand concepts that did not even occur to you before.

The CPU power of you mind is finite. It is easy to waste it all. Megatons break down the barriers.

All of your scripts should be **megatons**. 

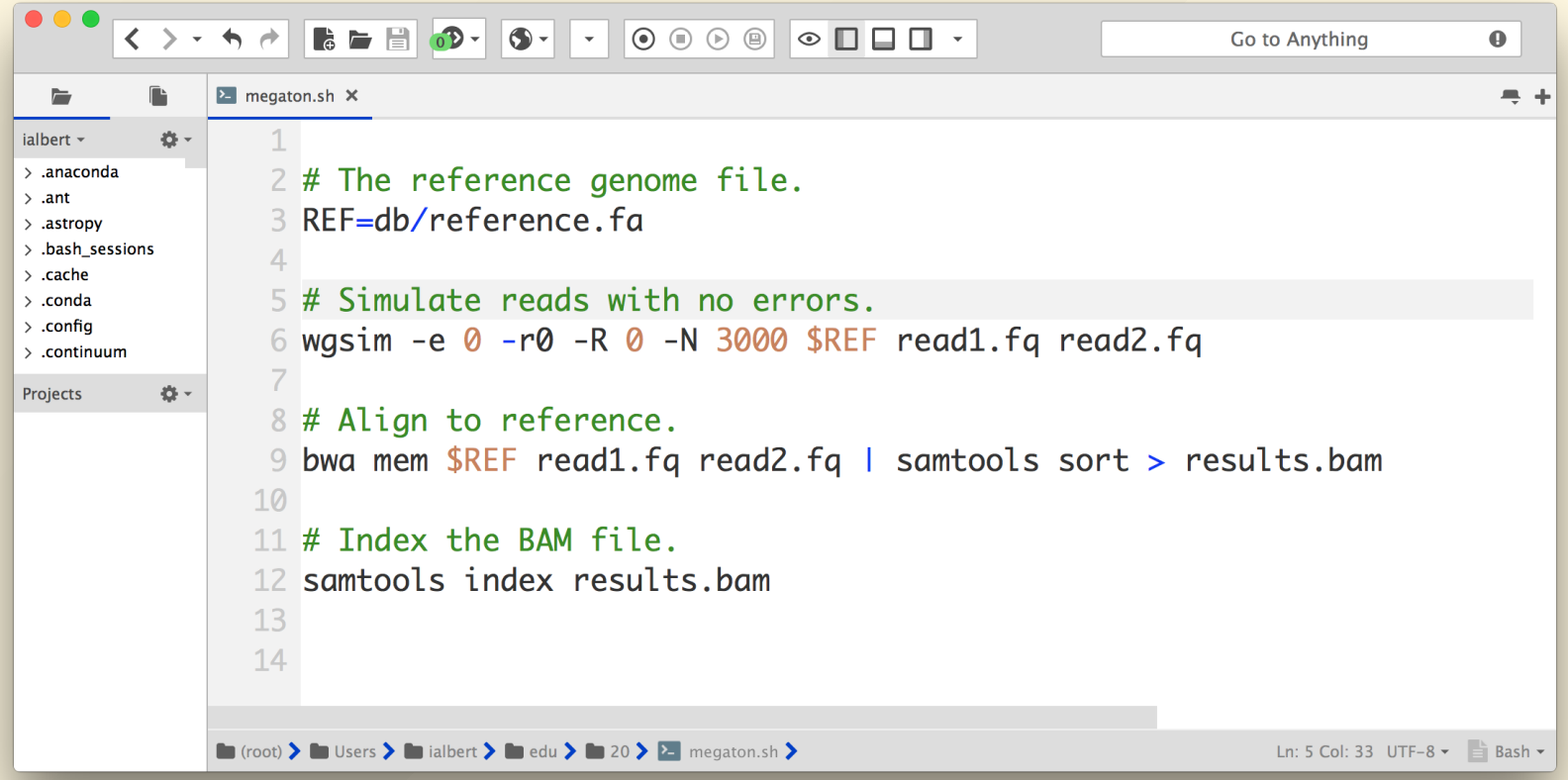
Build the simulation megaton

We want a megaton script that allows us to quickly visualize the effect of large scale genomic rearrangements.

We want to focus on the effect of changes and ignore everything else, alignment, BAM file, sorting, indexing ... etc

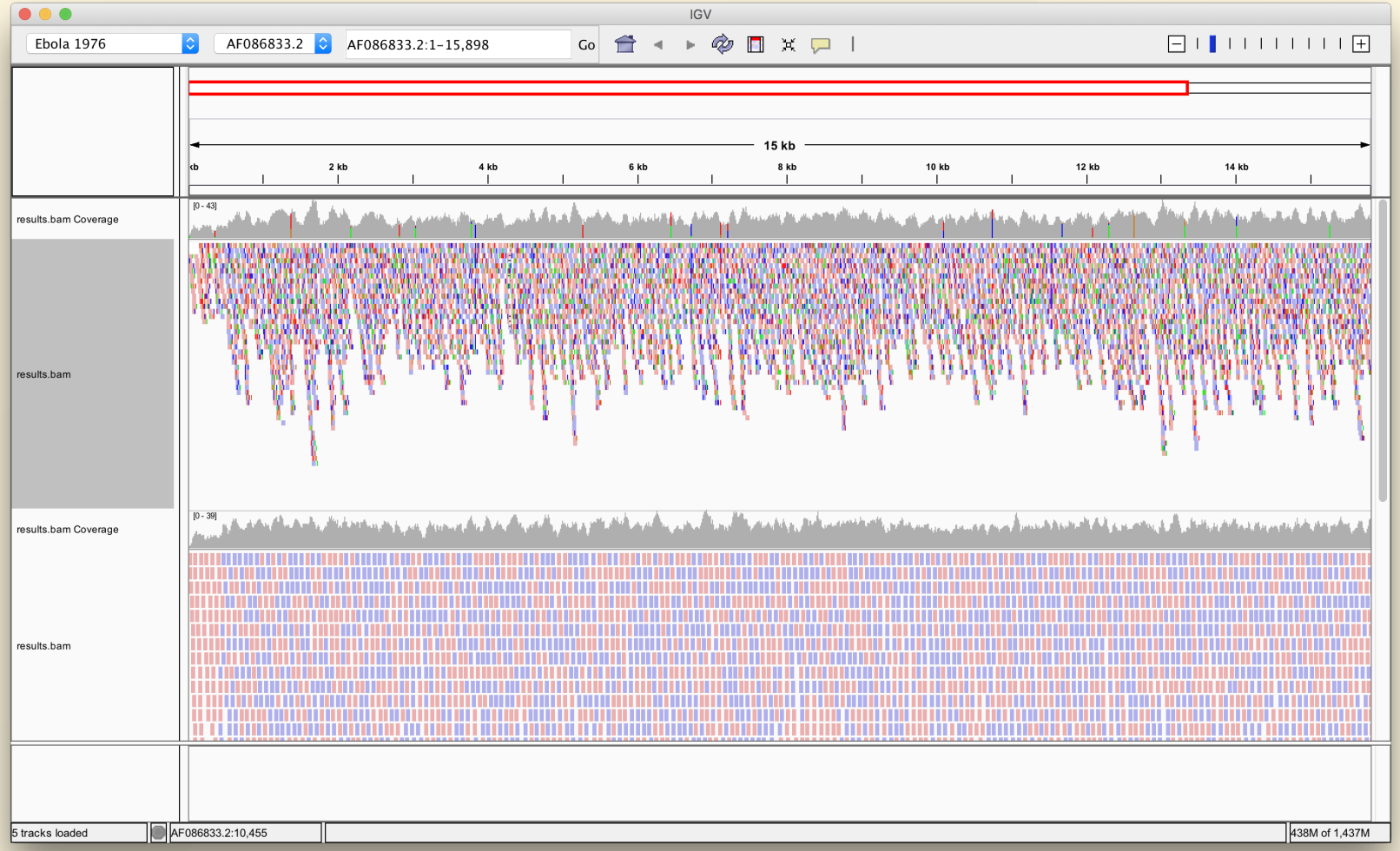
We want to just do it. 💪

Let's remove errors from simulation.




```
bash megaton.sh
```

Compare the two results.



Make the simulation use a different source.

Copy the reference to a file called genome.

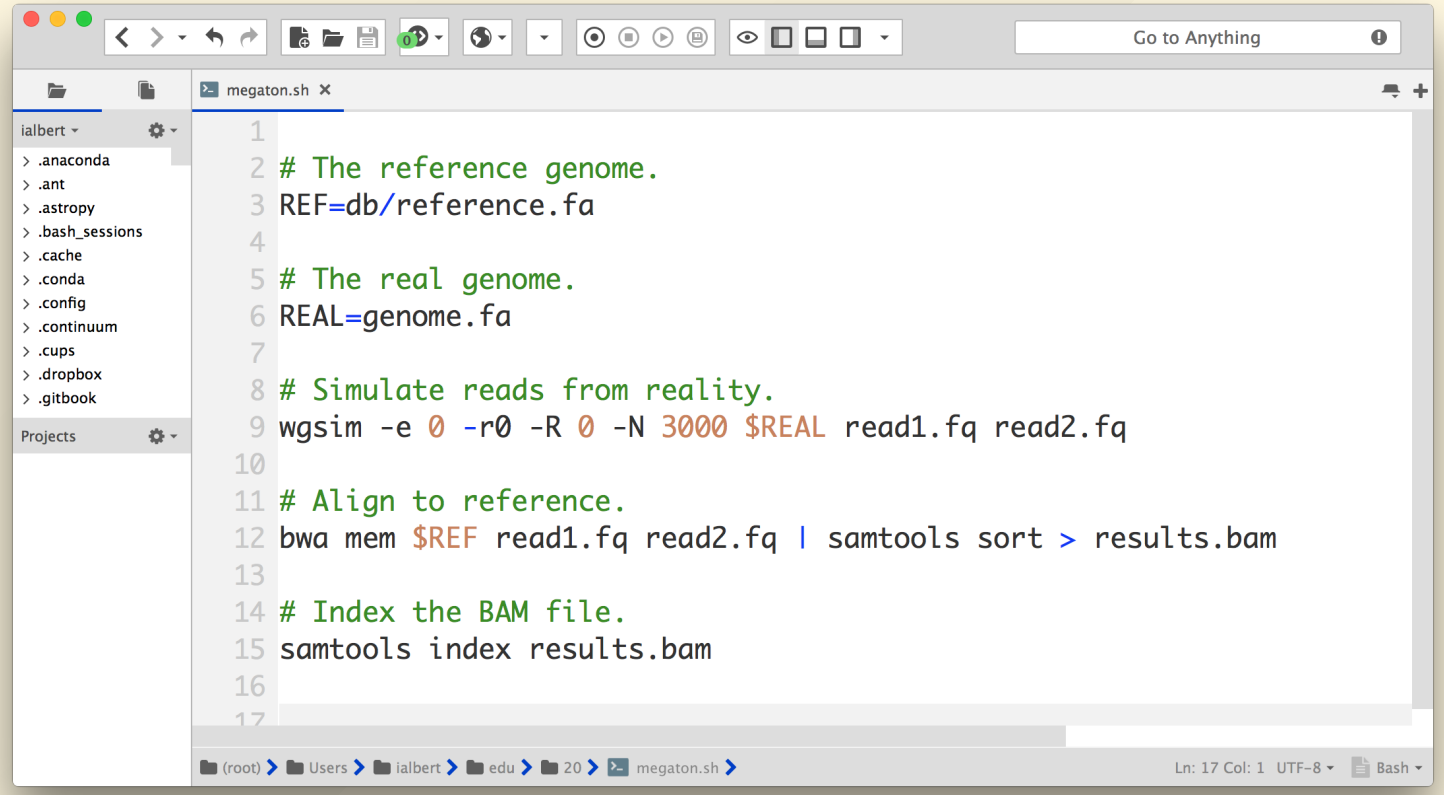
```
cp db/reference.fa genome.fa
```

We will now:

1. Modify the `genome.fa`
2. Simulate reads from `genome.fa`
3. Align against `db/reference.fa`

That is what data in real experiments look like.

Here is our megaton script



Modify the `genome.fa` - generate BAM in one step.

Reset genome

Any time you want to start fresh and "reset" the genome copy the reference over it.

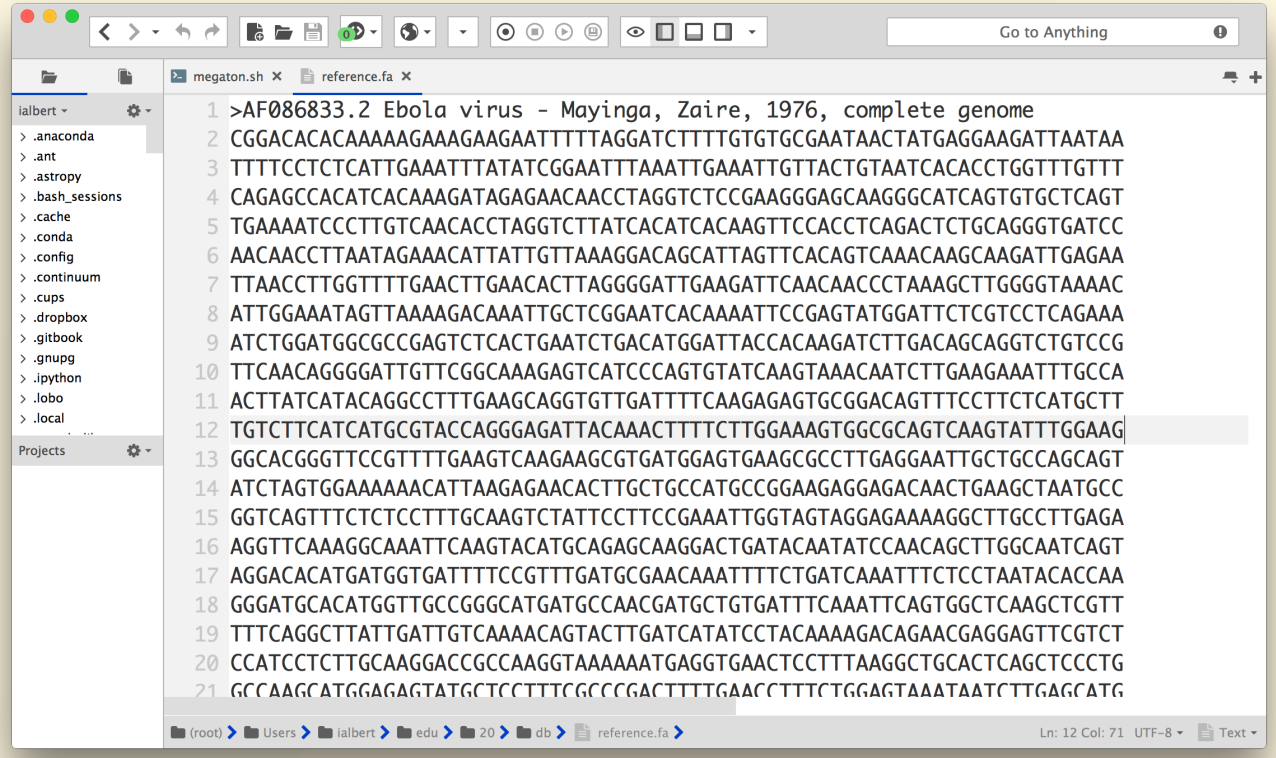
```
cp db/reference.fa genome.fa
```

We will modify `genome.fa` in various way, then run:

```
bash megaton.sh
```

then visualize the BAM file to see what it looks like.

Edit the genome, then rerun the megaton.



Edit your genome with an editor. Explore it that way.
We do it from command line to make it reproducible.

Copy number variation

The same region is present multiple times.

```
cat $REF | seqret --filter -sbegin 1 -send 2000 > start
cat start start start $REF | union -filter  > genome.fa
```

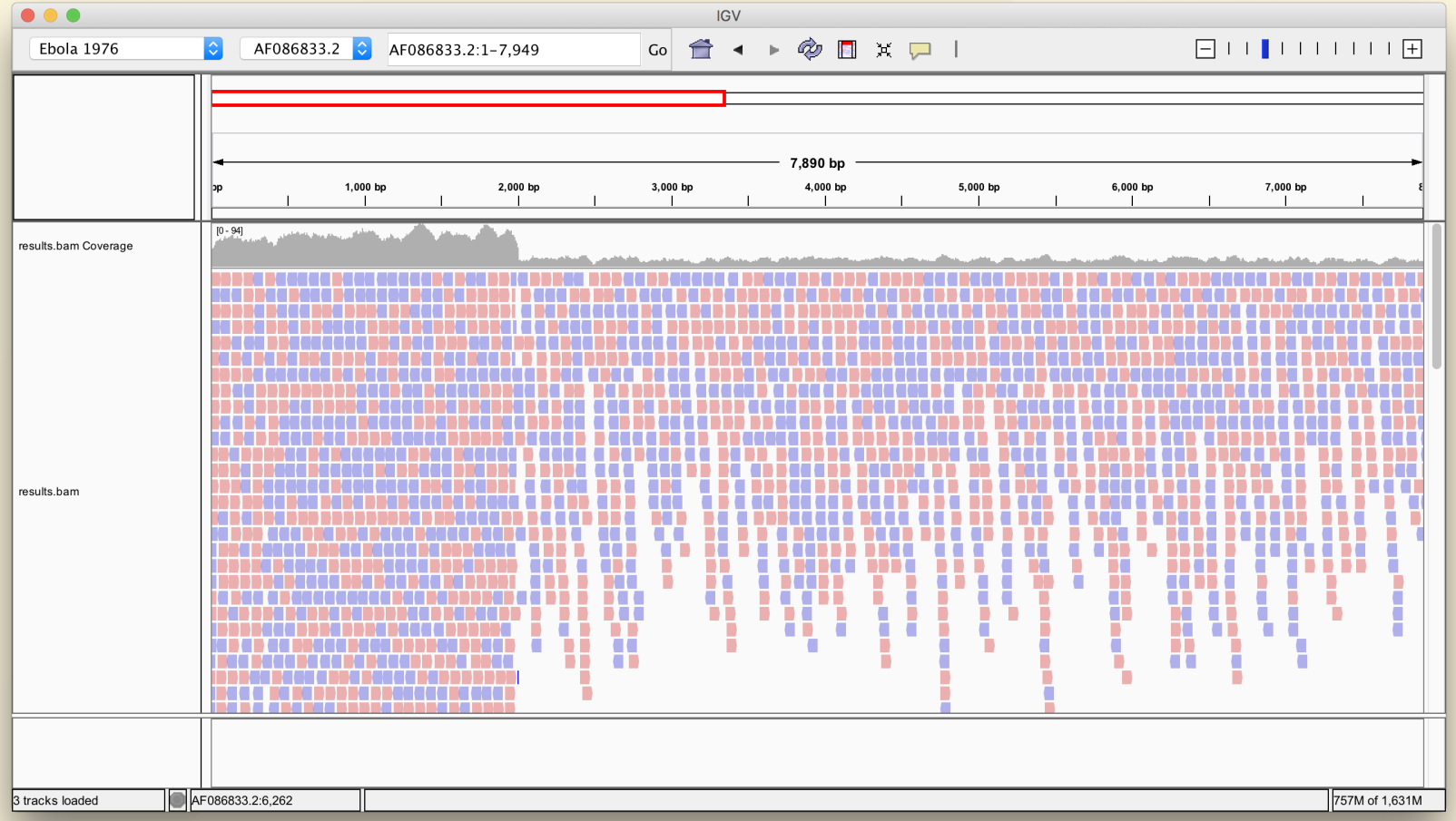
Reference has:

```
START-MIDDLE-END
```

Our real genome has:

```
START-START-START-MIDDLE-END
```

Copy number variations show up as coverage variation



Large scale deletion relative to reference

Easy to do with an editor. More complicated from command line

```
cat $REF | seqret --filter -sbegin 1 -send 2000 > start
cat $REF | seqret --filter -sbegin 3000 > end
cat start end | union -filter  > genome.fa
```

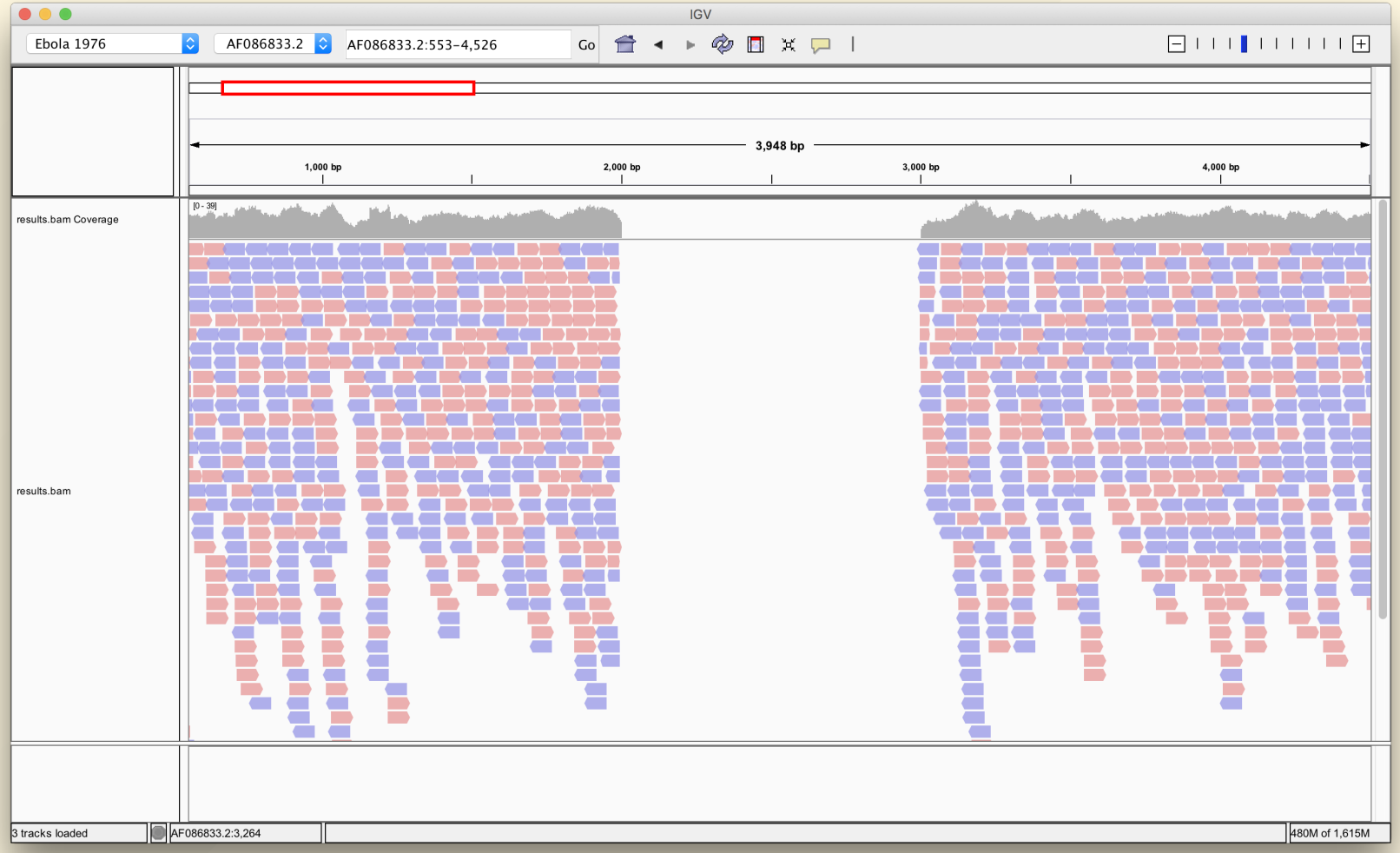
Reference has:

```
START-MIDDLE-END
```

Our real genome has:

```
START-END
```

Large scale deletion visualized



Large scale insertion relative to reference

This is also easy to do with an editor. But somewhat complicated from command line

```
cat $REF | seqret --filter -send 1000 > start
efetch -db nuccore -id NC_001802 -format fasta -seq_stop 1
cat $REF | seqret --filter -sbegin 1000 > end
cat start middle end | union -filter  > genome.fa
```

Reference has:

```
START-MIDDLE-END
```

Our real genome has:

```
START-SOMETHINGELSE-MIDDLE-END
```

What did we do?

. = Reference genome
x = Foreign sequence

We are matching fragments from

```
.................xxxxxxxx.................
```

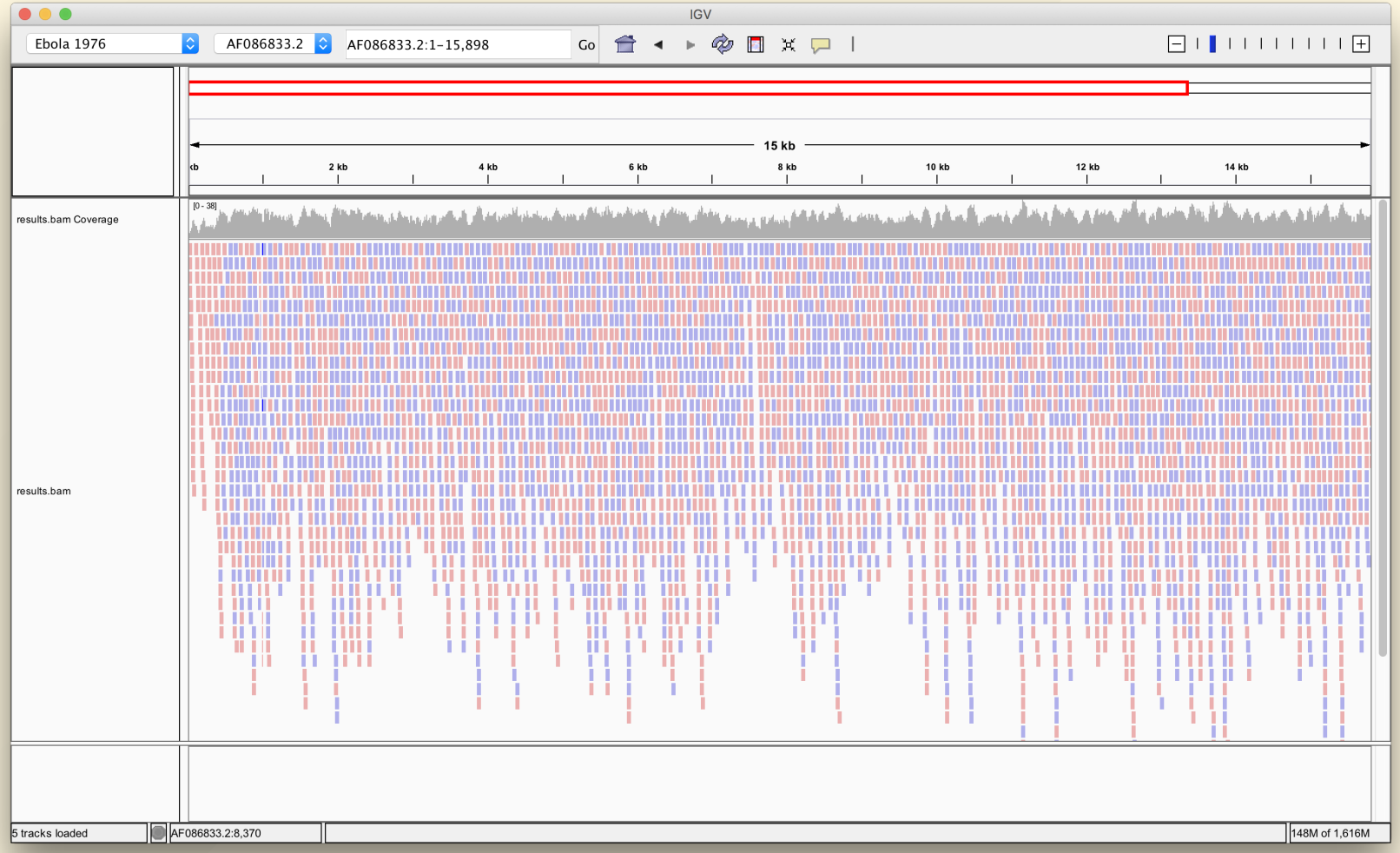
against

```
..................................
```

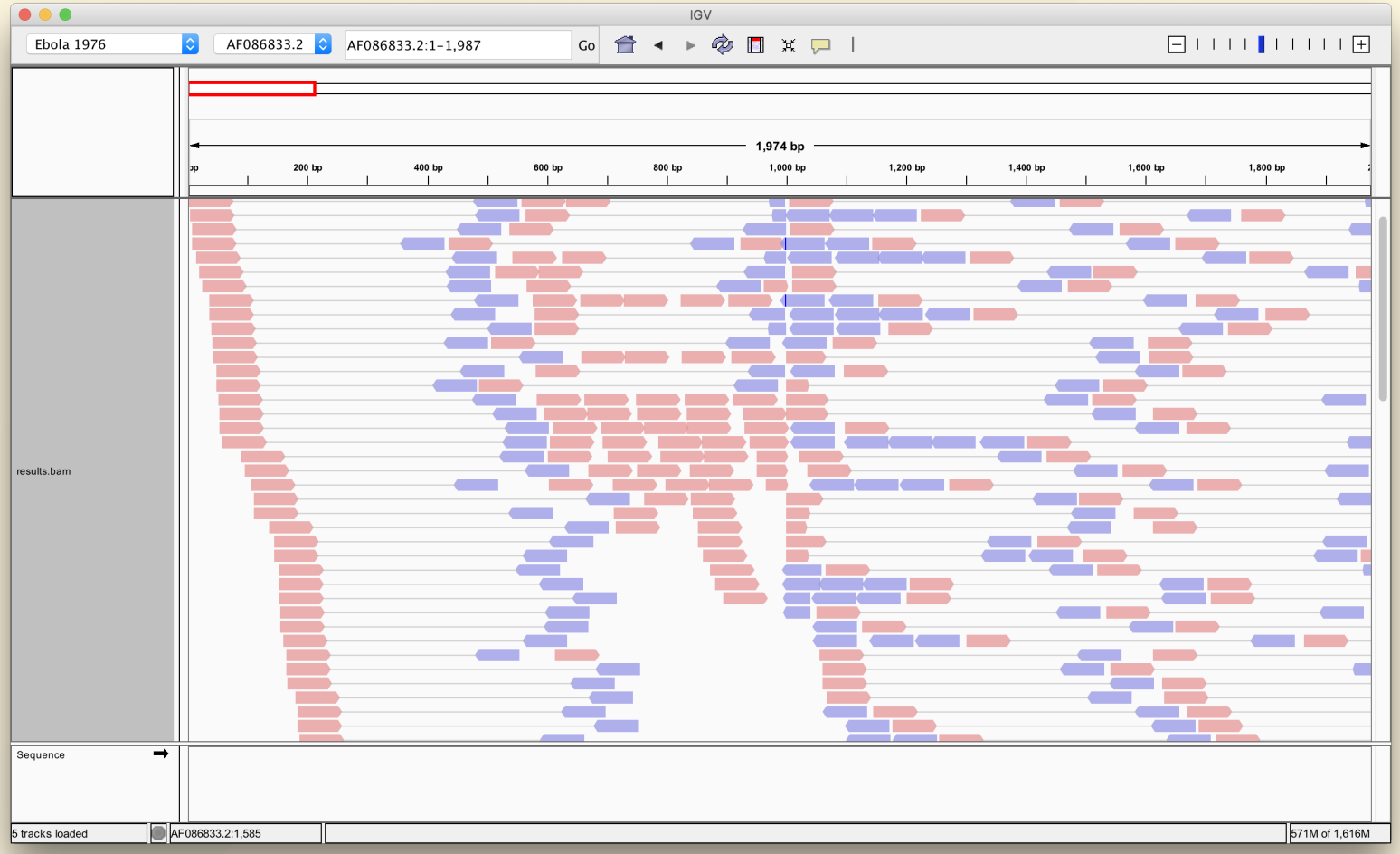
what happens with a read that contains?

```
....xxxx
```

Large scale insertion... Where is it?



"Orphan" reads



Reads that lost their mate (it is unmapped).

What if the insertion is shorter (200bp)?

Insert less than template lenght

```
cat $REF | seqret --filter -send 1000 > start
efetch -db nuccore -id NC_001802 -format fasta -seq_stop 2
cat $REF | seqret --filter -sbegin 1000 > end
cat start middle end | union -filter  > genome.fa
```

The results are even harder to see



No reads cross the border at 1000.

Insertions are more difficult to see

It is harder to detect something that you don't know is there.